

Secure-FEPRH

SECURE FACULTY OF ENGINEERING PROJECT & RESEARCH HUB
ENHANCED PROJECT PROPOSAL & ARCHITECTURE DOCUMENT

1. Introduction & Engineering Vision

The **Secure-FEPRH** is a proposed central, secure platform for managing academic projects and research at the Faculty of Engineering and Technology (Computer and Control Engineering Department).

It is designed as a software-only, enterprise-grade system that integrates Identity and Access Management (IAM) with a secure digital repository. The goal is to organize, document, and manage graduation projects using the latest cybersecurity techniques, Artificial Intelligence (Computer Vision and NLP), all while maintaining a zero-budget feasibility by leveraging existing hardware.

2. Problem Statement & Rationale

- **Lack of Digital Archiving & Protection:** Vital engineering resources (Python scripts, PIC, Assembly simulations, circuits) are often lost, leaked, or stolen due to the absence of a protected digital repository.
- **Repetition of Ideas & Lack of Originality:** The faculty struggles to instantly check proposed ideas to ensure they are not recycled from previous years.
- **Budget Limitations:** Purchasing dedicated hardware (sensors, smart gates, controllers) is financially unfeasible. A software-only solution leverages the existing infrastructure (PCs and built-in cameras) effectively.

3. Core Subsystems

- **IAM & Cyber Security Subsystem:**

- **Biometric Face MFA:** Uses Liveness Detection to prevent spoofing (e.g., using printed photos or phone screens) by ensuring the user in front of the camera is a live person.
- **Role-Based Access Control (RBAC):** Strict access matrix ensuring proper segregation of duties among Students, Supervisors, and the Department Head.

- **Smart Similarity Check (NLP):**

- Intelligent search engine that analyzes submitted abstracts and keywords.
- Compares submissions against a database of archived projects to generate a similarity report, aiding in the approval/rejection decision.

- **Secure Engineering Code Repository:**

- An encrypted digital archive integrated with the IAM system.
- Allows students to safely upload code and simulations. Access is strictly limited to authorized supervisors and the department head.

- **Smart Attendance:**

- Utilizes computer vision to automatically document the attendance of students and examiners during discussion sessions via the platform's camera.
- Generates immutable Audit Logs.

4. Extended Features (New Additions)

To elevate the platform to a comprehensive, enterprise-grade application, the following features have been integrated into the proposal:

A. Advanced Plagiarism Detection (AST/Code Similarity)

- **Description:** While NLP handles text (abstracts/reports), code requires specific analysis. Students can easily bypass NLP by renaming variables in copied code.
- **Implementation:** Integrate an Abstract Syntax Tree (AST) comparison tool or a service like **MOSS (Measure of Software Similarity)** to accurately detect code plagiarism across submissions.

B. Built-in Kanban & Task Management

- **Description:** Continuous evaluation is critical in engineering projects.

- **Implementation:** A Trello-like Kanban board (To Do, In Progress, Done) for each project group. Supervisors can track weekly sprint progress, assign tasks, and monitor individual student contributions, not just the final deliverable.

C. Basic Version Control Integration

- **Description:** Teach students industry standards by incorporating versioning.
- **Implementation:** Instead of simple file uploads, allow students to upload versions (v1.0, v1.1) or integrate webhooks to sync with private GitHub/GitLab repositories.

D. Automated Notifications & Alerts System

- **Description:** Proactive communication within the platform.
- **Implementation:** In-app and email notifications for approaching deadlines, new supervisor feedback on code, and critical security alerts (e.g., failed Face MFA attempts).

E. Department Head Analytics Dashboard

- **Description:** High-level overview for administrative decision-making.
- **Implementation:** Visual charts (via React libraries like Chart.js or Recharts) showing statistics such as popular programming languages per year, average similarity scores, and overall project domains.

5. Security & Technical Considerations

As a system focused on "Cyber Security" and "AI", the implementation must adhere to strict best practices to pass academic and security evaluations:

CAUTION: BIOMETRIC PRIVACY & STORAGE

- **Do NOT store raw photos** of students' faces in the database.
- **Do extract and store Face Encodings** (a 128-dimensional vector generated by Dlib/Face Recognition). This ensures that if the database is compromised, actual photos are safe and privacy is maintained.

WARNING: LIVENESS DETECTION FALLBACK

- Computer vision models can fail due to poor lighting, low-quality webcams, or students wearing masks/glasses.
- **Requirement:** Implement a secure fallback method. If Face MFA fails 3 consecutive times, trigger a Time-Based One-Time Password (TOTP) sent to the user's official university email.

IMPORTANT: ARABIC LANGUAGE SUPPORT IN NLP

- Project titles, proposals, and abstracts at Ibb University will likely be in Arabic.
- **Requirement:** Standard English NLP libraries (like NLTK's default stemmers) will be highly inaccurate. You must implement Arabic-specific NLP tools (e.g., CAMeL Tools, Farasa, or NLTK's ISRI Stemmer) for accurate tokenization, stop-word removal, and similarity checking.

IMPORTANT: ROBUST API SECURITY DEFENSES

- The backend (FastAPI) must be secured against common web vulnerabilities.
- **Rate Limiting:** Protect login endpoints against brute-force attacks.
- **Session Management:** Use short-lived JSON Web Tokens (JWT) for API authentication.
- **Input Sanitization:** Prevent SQL Injection and Cross-Site Scripting (XSS) on all text input fields.

TIP: FILE STORAGE & MALWARE PROTECTION

- With students uploading various simulation files and code, the server is at risk.
- **Requirement:** Implement strict file-type validation (allow only .py, .c, .cpp, .zip, .pdf, etc.).
- **Requirement:** Explicitly block executable formats (.exe, .bat, .sh) to prevent accidental or malicious malware execution on the university's servers.

6. System Architecture & Tech Stack Update

Layer	Technologies	Role in Project
Frontend (UI/UX)	React.js / Tailwind CSS	Responsive interfaces, camera integration, analytics dashboard, Kanban boards.
Backend (Core)	Python (FastAPI)	Secure APIs, RBAC logic, password encryption, file management.
AI & Vision	OpenCV / Dlib / Face_Recognition	Video stream capture, face encoding extraction, liveness detection.
NLP & Plagiarism	Scikit-Learn / Farasa / MOSS	Arabic text processing, TF-IDF vectorization, AST-based code similarity.
Database	PostgreSQL / MySQL	Secure storage of users, RBAC matrices, project metadata, and face embeddings.

7. Conclusion & Expected Value

The **Secure-FEPRH** platform transcends a simple project archive. With the addition of robust plagiarism detection, agile project management tools, and strict cybersecurity practices, it becomes a complete **Academic Project Lifecycle Manager**.

It perfectly aligns with the zero-budget constraint while providing a highly professional, scalable product that showcases the graduates' capabilities in AI, Web Development, and Cyber Security.